Final Project Scope: MLB Batting Lineup Comparison

1. Research Question

Make two big groups to determine if the inning(microgame) is using traditional vs OPS order

Groupings of traditonal and OPS order —--> Then sample t test to see which orders received more runs(resulted in better results)

Does a traditional-style batting order or an OPS-ranked batting order produce more expected runs per inning?

We will investigate this question using five MLB teams from the 2023-2026 regular seasons

- Los Angeles Dodgers
- Chicago Cubs
- St. Louis Cardinals
- Los Angeles Angels
- San Francisco Giants

Our objective is to determine whether rearranging the same hitters can improve expected offensive production.

2. Data

We will use three main sources of information:

Player statistics (2023-2026): OBP, OPS, SLG, SO, PA, and SO/PA to select players and construct batting orders.

Player outcome probabilities (2023-2026): Estimated probabilities of singles, doubles, triples, home runs, walks/HBP, and outs for each batter.

Play-by-play data (2023-2026): Approximately 30,500 offensive records across five teams, used to study baserunner advancement, outs, and run scoring.

3. Player Selection

For each team, we will select nine hitters based primarily on their 2023-2026 plate appearances.

We will prefer players with at least 150 PA, although exceptions may be necessary for teams affected by injuries or roster changes.

4. Batting Order Strategies

OBP - On Base Percentage
OPS - On Base Plus Slugging (Think of it as OBP + SLG)
SLG - Slugging (total bases divided by At Bats(AB))

Strategy A: Traditional-Style Lineup

The first five batting positions will be determined sequentially:

1. Highest OBP
2. Next highest OBP
3. Highest OPS among remaining hitters
4. Highest SLG among remaining hitters
5. Next highest OPS

Positions 6–9 will contain the remaining four hitters in a fixed order. (Descending OPS)

Strategy B: OPS-Ranked Lineup

The same five hitters selected for positions 1–5 will be reordered from highest to lowest OPS.

Positions 6–9 will remain identical to Strategy A. (Descending OPS)

This isolates the effect of rearranging the top five hitters.

Lineup Number	Traditional	Best to Worst
1	Highest OBP	Highest OPS
2	Next highest OBP	2nd Highest OPS
3	Highest OPS	3rd Highest OPS
4	Highest SLG	4th Highest OPS
5	Highest remaining OPS	5th Highest OPS

OPS, OBP,SLG —> 5- ….. Will be in descending order of OPS